7 **(a)** **(i)** max flux linkage = $64 \times 0.080 \times 0.71 \times 10^{-4}$ M1

$= 3.6 \times 10^{-4}$ Wb A1

(ii) E is zero from time 0 to time t

E has constant non-zero value between t and $2t$, B1
E opposite sign between $2t$ and $4t$ B1

(b) decrease in flux density of solenoid (towards end) results in decrease in flux linkage in coil B1

According to Faradays' Law, induced emf in coil leads to (induced) current because Coil C is connected B1

which causes magnetic field around coil that interact with solenoid magnetic field to create attractive force B1